# Question-10.3.25

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# Question

Find equation of all lines having slope 2 which are tangents to curve $y + \frac{2}{x-3} = 0$

# Solution

Given curve:

$$y + \frac{2}{x-3} = 0 \implies yx - 3y + 2 = 0 \tag{1}$$

compare with

$$\left(x\right)^T \mathbf{V} \left(x\right) + 2\left(u\right)^T \left(x\right) + f = 0 \tag{2}$$

$$\mathbf{V} = \begin{pmatrix} 0 & 0.5 \\ 0.5 & 0 \end{pmatrix}, \left(u\right) = \begin{pmatrix} 0 \\ -1.5 \end{pmatrix}, f = 2 \tag{3}$$

From above,the normal vector is

$$\left(n\right) = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \tag{4}$$

# Solution

Points of contact:

$$\left(q\right)_i = \mathbf{V}^{-1}(\kappa_i \left(n\right) - \left(u\right)), i = 1, 2 \tag{5}$$

$$\mathbf{V}^{-1} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \left(u\right)^T \mathbf{V}^{-1} \left(u\right) = 0, f = 2, \left(n\right)^T \mathbf{V}^{-1} \left(n\right) = -8 \tag{6}$$

$$\kappa_i = \pm\sqrt{\frac{\left(u\right)^T \mathbf{V}^{-1} \left(u\right) - f}{\left(n\right)^T \mathbf{V}^{-1} \left(n\right)}} = \pm\sqrt{\frac{-2}{-8}} = \pm 0.5 \tag{7}$$

For $\kappa_1 = 0.5$:

$$\left(q\right)_1 = \mathbf{V}^{-1}(0.5 \left(n\right) - \left(u\right)) = \mathbf{V}^{-1} \begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \tag{8}$$

For $\kappa_2 = -0.5$:

$$\left(q\right)_2 = \mathbf{V}^{-1}(-0.5 \left(n\right) - \left(u\right)) = \mathbf{V}^{-1} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \tag{9}$$

# Solution

Tangent equation at contact point:

$\left(\mathbf{V}\left(q\right)+\left(u\right)\right)^{T}\left(x\right)+\left(u\right)^{T}\left(q\right)+f=0$

For $\left(q\right)_1=\begin{pmatrix}4\\-2\end{pmatrix}$:

$$\mathbf{V}\left(q\right)_1+\left(u\right)=\begin{pmatrix}-1\\2\end{pmatrix}+\begin{pmatrix}0\\-1.5\end{pmatrix}=\begin{pmatrix}-1\\0.5\end{pmatrix} \tag{10}$$

$$\left(u\right)^{T}\left(q\right)_1=3 \tag{11}$$

For $\left(q\right)_2=\begin{pmatrix}2\\2\end{pmatrix}$:

$$\mathbf{V}\left(q\right)_2+\left(u\right)=\begin{pmatrix}1\\1\end{pmatrix}+\begin{pmatrix}0\\-1.5\end{pmatrix}=\begin{pmatrix}1\\-0.5\end{pmatrix}\left(u\right)^{T}\left(q\right)_2=-3 \tag{12}$$

The equations of tangents are

$$\begin{pmatrix}-1 & 0.5\end{pmatrix}\mathbf{x}+5=0 \tag{13}$$

$$\begin{pmatrix}1 & -0.5\end{pmatrix}\mathbf{x}-1=0 \tag{14}$$

# graph

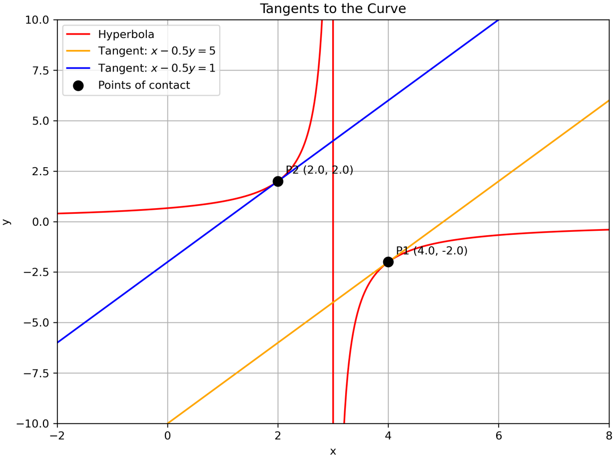


Figure:

# C-Code

```c
// code.c
#include <stdio.h>

void tangent_points(double* pts) {
    // Store two points of tangency (x1, y1, x2, y2)
    pts[0] = 4.0;
    pts[1] = -2.0;
    pts[2] = 2.0;
    pts[3] = 2.0;
}
```

# Python-Code

```
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

lib = ctypes.CDLL('./points.so')

PointsArr = ctypes.c_double * 4
pts = PointsArr()

lib.tangent_points(pts) # Fill pts with tangent points

points = np.array([pts[0], pts[1], pts[2], pts[3]]).reshape(2, 2)
```

```
# plot.py
import numpy as np
import matplotlib.pyplot as plt
from call import points

# Hyperbola: y = -2/(x-3)
x = np.linspace(-2, 8, 400)
with np.errstate(divide='ignore'):
    y = -2 / (x - 3)

valid = np.abs(x - 3) > 1e-8 # exclude singularity
x_h = x[valid]
y_h = y[valid]
fig, ax = plt.subplots(figsize=(8, 6))
# Plot hyperbola
ax.plot(x_h, y_h, 'r', label='Hyperbola')
# Tangents: x - 0.5 y = 5 -> y = 2x - 10, x - 0.5 y = 1 -> y = 2x
    - 2
x_vals = np.linspace(-2, 8, 400)
```

# Python-Code

```
ax.plot(x_vals,2*x_vals -10,'orange',label='Tangent:$x -0.5y =5$')
ax.plot(x_vals,2*x_vals -2,'blue',label='Tangent:$x -0.5y =1$')
# Points of contact from C
ax.scatter(points[:, 0], points[:, 1], color='black', zorder=5, s=75, label='Points of contact')
for i, (x_c, y_c) in enumerate(points):
    ax.annotate(f'P{i+1} ({x_c:.1f}, {y_c:.1f})', (x_c, y_c), textcoords='offset points', xytext=(7,7), ha='left')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Tangents to the Curve')
ax.set_xlim(-2, 8)
ax.set_ylim(-10, 10)
ax.grid(True)
ax.legend()
plt.tight_layout()
plt.savefig('tangents.png', dpi=300)
plt.show()
```